GRUPO FINANCIERO BANORTE

MANUAL DE USUARIO

Clave del Documento

Procedimiento para Cifrado y Decifrado simétrico / asimétrico. Ejemplo de componente en Java

Responsables de la Aplicación

Dominio	Subdominio
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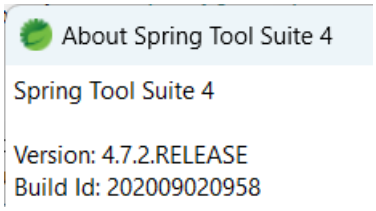
Control de Versiones del Documento

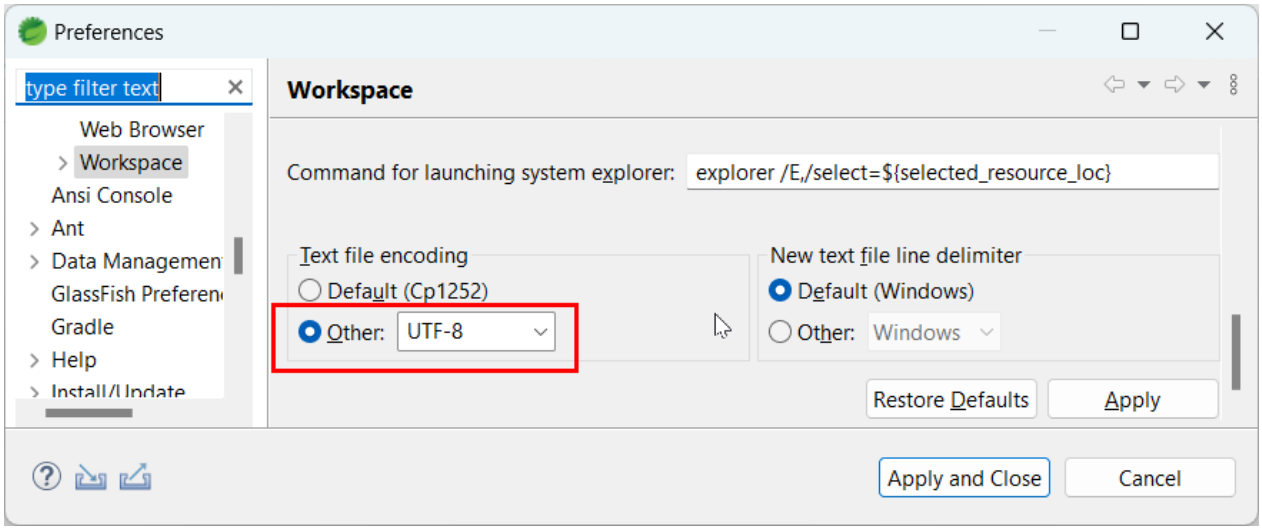
Fecha Modificación	Versión	Autor

Introducción

El presente documento tiene la finalidad de proporcionar al usuario del comercio el conocimiento necesario para poder cifrar y descifrar la información (mediante los Algoritmos provistos por Banorte) que se procesa en el aplicativo VCE (Ventana de Comercios).

El presente documento detalla un ejemplo de implementación con el lenguaje de programación Java en Spring Tools Suite 4.7.2.RELEASE.





Lenguaje de desarrollo.	Java 1.8
Framework.	SpringBoot 2.7.1
Herramienta de desarrollo.	Spring Tools Suite 4.7.2.RELEASE
Tipo de proyecto	Maven
Referencias adicionales.	spring-boot-starter spring-boot-starter spring-boot-starter-log4j spring-boot-starter-tomcat spring-boot-starter-logging commons-codec

Objetivo.

Obtener cadena de datos para invocar a VCE, se requiere armar una cadena compuesta por dos subcadenas separadas por el delimitador ":::".

Subcadena1:::Subcadena2

Procedimiento para generar Subcadena1:

Significado Subcadena1.

- Cadena cifrada de forma simétrica con RSA utilizando el certificado proporcionado por Banorte.
- El contenido de la cadena son tres elementos generados al momento de cifrar **Subacdena2** separados por el delimitador "::"
  - **ivHex::saltHex::passprhase**
- Algoritmo utilizado: RSA/ECB/OAEPWITHSHA-256ANDMGF1PADDING
- Del certificado proporcionado por Banorte se tiene que extraer la llave pública del certificado convirtiendo el certificado de PEM a x.509.

Código:

```java
package com.cifrado.utils;

import java.io.ByteArrayInputStream;
import java.io.InputStream;
import java.nio.charset.StandardCharsets;
import java.security.InvalidKeyException;
import java.security.KeyFactory;
import java.security.NoSuchAlgorithmException;
import java.security.NoSuchProviderException;
import java.security.PublicKey;
import java.security.cert.CertificateFactory;
import java.security.cert.X509Certificate;
import java.security.spec.X509EncodedKeySpec;

import javax.crypto.BadPaddingException;
import javax.crypto.Cipher;
import javax.crypto.IllegalBlockSizeException;
import javax.crypto.NoSuchPaddingException;
import org.apache.commons.codec.binary.Base64;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class CifradoRSA
{
  private static Logger logger = LoggerFactory.getLogger(com.cifrado.utils.CifradoRSA.class);

  public static String encrypt(String plainText, String publicKey) {
    try {
      Cipher encryptCipher = Cipher.getInstance("RSA/ECB/OAEPWITHSHA-256ANDMGF1PADDING", "SunJCE");


      byte[] b64decoded = Base64.decodeBase64(publicKey);
      CertificateFactory certFactory = CertificateFactory.getInstance("X.509");
      InputStream in = new ByteArrayInputStream(b64decoded);
      X509Certificate certificate = (X509Certificate)certFactory.generateCertificate(in);

      encryptCipher.init(1, certificate.getPublicKey());

      byte[] encryptedData = encryptCipher.doFinal(plainText.getBytes(StandardCharsets.UTF_8));
      String encodeToString = Base64.encodeBase64String(encryptedData);

      logger.debug("RSA Encrypt Provider: [{}], [{}]", new Object[] { encryptCipher.getProvider().getName(), encryptCipher.getAlgorithm() });
      return encodeToString;
    } catch (InvalidKeyException e) {
      logger.error("Llave RSA invalida", e);
    } catch (NoSuchAlgorithmException e) {
      logger.error("No es posible utilizar el algoritmo especificado RSA", e);
    } catch (NoSuchProviderException e) {
      logger.error("No es posible Localizar el proveedor RSA especificado", e);
    } catch (NoSuchPaddingException e) {
      logger.error("No es posible Localizar el padding especificado", e);
    } catch (IllegalBlockSizeException e) {
      logger.error("Tamaño de bloque ilegal", e);
    } catch (BadPaddingException e) {
      logger.error("Padding incorrecto", e);
    } catch (Exception e) {
      e.printStackTrace();
    }
    return null;
  }
}
```

plainText = Cadena **ivHex::saltHex::passprhase**

Ej.

7434316130317A753068723272746569::3076356C346F7366387675337A316171::g-T=Kq81GSgxRhc%

rsaPublicKey = Cadena que contiene el certificado proporcionado por Banorte

Ej.

MIIGdDCCBVygAwIBAgITFwAAxu3koN5OHp90hgABAADG7TANBgkqhkiG9w0BAQsFADB5MQswCQ
YDVQQGEwJNWDEZMBcGCgmSJomT8ixkARkWCWdmYmFub3J0ZTEeMBwGA1UEChMVU2VndXJp
ZGFkIGluZm9ybWF0aWNhMS8wLQYDVQQDEyZHcnVwbyBGaW5hbmNpZXJvIEJhbm9ydGUgQ0EgR
W50ZXJwcmlzZTAeFw0yMjAxMjExNzU3NDFaFw0yMzAxMjExNzU3NDFaMIGsMQswCQYDVQQGEwJ
NWDELMAkGA1UECBMCREYxDzANBgNVBAcTBk1leGljbzErMCkGA1UEChMiR3J1cG8gRmluYW5ja
WVybyBCYW5vcnRlIFNBQiBkZSBDVjEkMCIGA1UECxMbU2VndXJpZGFkIGRlIGxhIEluZm9ybWFjaW9
uMSwwKgYDVQQDEyNtdWx0aWNvYnJvc2tleS5kZXYudW5peC5iYW5vcnRlLmNvbTCCASIwDQYJKo
ZIhvcNAQEBBQADggEPADCCAQoCggEBAOW0T9LRmHPdL0gHFLojhObdEPnHbiQ0D84O3v96lZWH
dW19g8ZC772rMCkExT47qANVhCTEgnFhz80XOiMUSypODPWVkfcpgts13Lfxq07ltMUTK69M6fcVXjR
GBKUs3Gcv0kVYcCgKf2+q2jxTQeYRUraAzu6umZY5W98lJzqryED4Du3pNwla6uj5+X1g0Uds8oXSj4Ml
PjPUHedQ0xa/3GES+vMVCYB1qDnEwhmZFnLnzjCN+VM7mNGDP4kbGEYY+4asetd1QT2SIxmXIpnG
XWNHYeqNZoN3XerM0zVDQ/vgWkyLocSoocIxl1Bpn6hp/ycO0sTzA97FYUk5eJECAwEAAaOCAr8wgg
K7MC4GA1UdEQQnMCWCI211bHRpY29icm9za2V5LmRldi51bml4LmJhbm9ydGUuY29tMB0GA1UdD
gQWBBQBtoMxQxbByCO1tuY3s9yEFxpQ2TAfBgNVHSMEGDAWgBTQBPbPr59ruX3bXiA0YK0zB6UM
0zCB7QYDVR0fBIHlMIHiMIHfoIHcoIHZhoHWbGRhcDovLy9DTj1HcnVwbyUyMEZpbmFuY2llcm8lMjBC
YW5vcnRlJTIwQ0ElMjBFbnRlcnByaXNlLENOPUNBLUJBTk9SVEUtUkEsQ049Q0RQLENOPVB1Ymxp
YyUyMEtleSUyMFNlcnZpY2VzLENOPVNlcnZpY2VzLENOPUNvbmZpZ3VyYXRpb24sREM9Z2ZiYW5v
cnRlP2NlcnRpZmljYXRlUmV2b2NhdGlvbkxpc3Q/YmFzZT9vYmplY3RDbGFzcz1jUkxExN0cmlidXRpb
25Qb2ludDCB2wYIKwYBBQUHAQEEgc4wgcswgcgGCCsGAQUFBzAChoG7bGRhcDovLy9DTj1HcnV
wbyUyMEZpbmFuY2llcm8lMjBCYW5vcnRlJTIwQ0ElMjBFbnRlcnByaXNlLENOPUFJQSxDTj1QdWJsaW
MlMjBLZXklMjBTZXJ2aWNlcyxDTj1TZXJ2aWNlcyxDTj1Db25maWd1cmF0aW9uLERDPWdmYmFub3J
0ZT9jQUNlcnRpZmljYXRlP2Jhc2U/b2JqZWN0Q2xhc3M9Y2VydGlmaWNhdGlvbkF1dGhvcml0eTALBg
NVHQ8EBAMCBaAwPAYJKwYBBAGCNxUHBC8wLQYlKwYBBAGCNxUIg+v5BoK1j0eNkROCp58Ggu
HddCKCxphfgtj9DwIBZAIBDzATBgNVHSUEDDAKBggrBgEFBQcDATAbBgkrBgEEAYI3FQoEDjAMMA
oGCCsGAQUFBwMBMA0GCSqGSIb3DQEBCwUAA4IBAQBi1NI4gHDbgldA62xFDtt/9nsSMtPZ3yCi2E
vX+F8OnhecGDvq2i3/87gfTr9K0nWzA/3EwM21i6/JlcIyAJaKQQwu551Ic7ky5OBUw811be0MltwzvsdOA
+Nd641+Fh+VfW5/yaIQccXF2Zm7MBrCoXbDgzoX+leeIeqq55orBUqdA8/woYC796+ERHdQqpuVuowit
gnOZDFwr0Dm2kg02ZYHwUp5Pf55MfVYrT1AgVkP7X4KH1EcjNCRyYnOuVbUSzVtZSBYLCkhgB9Baf
YSvd78M1yS8Uc2y1wKh4D2TU/UpT3WXjbpA9P8Cn39WxX3EUfhrOwcUOtmkHA45Hcf

Resultado (el resultado es la **Subcadena1)**:

Ej.

X/GqsF19cIXMxzqlLCQWO/ZySIlL16JwJNiKSmS5pF8YjUbdK6a9PFpDTtafxdmKVyQG6unYSRCy0Gim
lyD+1jFvfgfYYF14Z1DtZU4Dq5ZOVhGXnK4DXUJhHM2T1dv3EtwFSurVMKGaBi6wGEX1W6xS90PvFP
Yv9x5NDnvkIGUvZmfE1HkP6pMUTZoxGFD1caLW13eSrsMkZ6JfeVo6kDUudKdiOMWrXFuHf7CT9Sw
9krq2Hwm8uUodIKMbGsXB5XOBZZqDAxeOsTaO78+DxgRnHtB+c2J3e0PuyJmZR0/PNs7p63gmUh44
yxlIJwFydQULCUAqGrPaONgTIgiHGA==

Procedimiento para generar Subcadena2:

Significado Subcadena2:

- Cadena cifrada de forma asimétrica con AES utilizando llaves generadas de manera dinámica.
- El contenido son los datos en formato JSON requeridos por VCE.
- Algoritmo utilizado: AES/CTR/NoPadding
- Se utilizan 4 elementos para lograr el cifrado:
  - **passPrhase**
    - Cadena de 16 caracteres.
    - Se generan 16 caracteres de manera aleatoria de una cadena predefinida:
      - 0123456789ABCDEFGHIJKLMNOPQRSTUVWXTZabcdefghiklmnopqrstuvwxyz*&-%/!?*+=()
  - **ivHex**
    - Cadena de 32 caracteres
    - Se generan 16 caracteres de manera aleatoria de una cadena predefinida:
      - 0123456789abcdefghiklmnopqrstuvwxyz
    - De los 16 caracteres se obtiene su representación en hexadecimal de cada caracter para obtener la cadena final de 32.
  - **saltHex**
    - Cadena de 32 caracteres
    - Se generan 16 caracteres de manera aleatoria de una cadena predefinida:
      0123456789abcdefghiklmnopqrstuvwxyz
    - De los 16 caracteres se obtiene su representación en hexadecimal de cada caracter para obtener la cadena final de 32.
  - **key**
    - Llave generada mediante el método Rfc2898DeriveBytes utilizando el algoritmo HashAlgorithmName.SHA1 y los elementos **saltHex** (los 16 caracteres convertidos en un arreglo de bytes) y **passprhase**.
    - De la llave generada se obtienen los primeros 16 bytes.

Código:

Generación datos aleatorios:

```
CifradoAES cifradoaes = CifradoAES.getInstance();
String ivHex = CifradoAES.createInitializationVector();
String saltHex = CifradoAES.createSalt();
String passPhrase = CifradoAES.createRandomKey();
```

```
public static com.cifrado.utils.CifradoAES getInstance() {
  if (instance == null) {
    instance = new com.cifrado.utils.CifradoAES(128, 1000);
  }
  return instance;
}

private CifradoAES(int keySize, int iterationCount) {
  this.keySize = keySize;
  this.iterationCount = iterationCount;
  try {
    this.cipher = Cipher.getInstance("AES/CTR/NoPadding");
  }
  catch (NoSuchAlgorithmException|javax.crypto.NoSuchPaddingException e) {
    throw fail(e);
  }
}


public static String createInitializationVector() {
  String cadenaRandom = "0123456789abcdefghiklmnopqrstuvwxyz";
  StringBuilder iv = new StringBuilder();
  Random rnd = new Random();
  while (iv.length() < 16) {
    int index = (int)(rnd.nextFloat() * cadenaRandom.length());
    iv.append(cadenaRandom.charAt(index));
  }
  char[] ivString = Hex.encodeHex(iv.toString().getBytes(StandardCharsets.UTF_8));
  return String.valueOf(ivString);
}


public static String createSalt() {
  String ramdom = "0123456789abcdefghiklmnopqrstuvwxyz";
  StringBuilder salt = new StringBuilder();
  Random rnd = new Random();
  while (salt.length() < 16) {
    int index = (int)(rnd.nextFloat() * ramdom.length());
    salt.append(ramdom.charAt(index));
  }
  char[] saltString = Hex.encodeHex(salt.toString().getBytes(StandardCharsets.UTF_8));
  return String.valueOf(saltString);
}
```

org.apache.commons.codec.binary.Hex

Converts hexadecimal Strings. The charset used for certain operation can be set, the default is set in `DEFAULT_CHARSET_NAME` This class is thread-safe.

Since:
1.1

Version:
$Id: Hex.java 1619948 2014-08-22 22:53:55Z ggregory $

```
public static String createRandomKey() {
  String ramdomkey = "0123456789ABCDEFGHIJKLMNOPQRSTUVWXTZabcdefghiklmnopqrstuvwxyz*&-%/!?*+=()";
  StringBuilder keyRnd = new StringBuilder();
  Random rnd = new Random();
  while (keyRnd.length() < 16) {
    int index = (int)(rnd.nextFloat() * ramdomkey.length());
    keyRnd.append(ramdomkey.charAt(index));
  }
  return keyRnd.toString();
}
```

Invocación al método de cifrado:

```
String cadenaCifradaAES = cifradoaes.encrypt(saltHex, ivHex, passPhrase, cadenaACifrarDecodificada);
```

```
public String encrypt(String salt, String iv, String passphrase, String plaintext) {
  SecretKey key = generateKey(salt, passphrase);
  byte[] encrypted = doFinal(1, key, iv, plaintext.getBytes(StandardCharsets.UTF_8));
  return base64(encrypted);
}

private byte[] doFinal(int encryptMode, SecretKey key, String iv, byte[] bytes) {
  try {
    this.cipher.init(encryptMode, key, new IvParameterSpec(hex(iv)));
    return this.cipher.doFinal(bytes);
  }
  catch (InvalidKeyException|java.security.InvalidAlgorithmParameterException|javax.crypto.IllegalBlockSizeException|javax.crypto.BadPaddingException e) {
    return null;
  }
}

private SecretKey generateKey(String salt, String passphrase) {
  try {
    SecretKeyFactory factory = SecretKeyFactory.getInstance("PBKDF2WithHmacSHA1");
    KeySpec spec = new PBEKeySpec(passphrase.toCharArray(), hex(salt), this.iterationCount, this.keySize);
    SecretKey key = new SecretKeySpec(factory.generateSecret(spec).getEncoded(), "AES");

    return key;
  }
  catch (NoSuchAlgorithmException|java.security.spec.InvalidKeySpecException e) {
    return null;
  }
}
```

cadenaACifrarDecodificada = Cadena texto plano que contiene el JSON de los datos requeridos por VCE.

Ej.

{"merchantId":"**********","name":"**********","password":"********","mode":"AUT","controlNumber":"PBA22042709695XXFA36H01","terminalId":"12348","amount":"1.00","merchantName":"Pruebas Banorte","merchantCity":"Monterrey","lang":"ES","customerRef1":"22042709695XXFA36H01","customerRef2":"","customerRef3":"","customerRef4":"","customerRef5":"","emailCdmx":"correo@correo.group","phoneCdmx":"5566778899","conceptoCdmx":"|Concepto:|Impuestos sobre Tenencia de Veh?culos y derechos|Referencia:|P31BFM|Ejercicio Fiscal:|2021","billToFirstName":"Nombre Cliente","billToLastName":"Apellidos Cliente","billToStreet":"Eugenia","billToStreetNumber":"197","billToStreetNumbe2":"","billToStreet2Col":"Narvarte","billToStreet2Del":"Benito Juarez","billToCity":"CDMX","billToState":"DF","billToCountry":"MX","billToPhoneNumber":"5566778899","billToPostalCode":"03999","billToEmail":"review@banorte.com","billToCustomerId":"","billToCustomerPassword":"","billToDateOfBirth":"","billToHostname":"LSTK","billToHttpBrowserEmail":"","billToIpAddress":"127.0.0.1","comments":"Compra a MSI","shipToFirstName":"","shipToLastName":"","shipToStreetNumber":"","shipToStreetNumber2":"","shipToStreet2Col":"","shipToStreet2Del":"","shipToCity":"","shipToState":"","shipToCountry":"","shipToPostalCode":"","shipToPhoneNumber":"","merchantDefinedDataField3":"","merchantDefinedDataField4":"","merchantDefinedDataField5":"","merchantDefinedDataField8":"","merchantDefinedDataField6":"","merchantDefinedDataField7":"","merchantDefinedDataField9":"","merchantDefinedDataField10":"","merchantDefinedDataField11":"","merchantDefinedDataField12":"","merchantDefinedDataField13":"","merchantDefinedDataField14":"","merchantDefinedDataField15":"","merchantDefinedDataField16":"","merchantDefinedDataField17":"","merchantDefinedDataField18":"","merchantDefinedDataField19":"","merchantDefinedDataField20":"","merchantDefinedDataField21":"","merchantDefinedDataField22":"","merchantDefinedDataField23":"","merchantDefinedDataField24":"","merchantDefinedDataField25":"","merchantDefinedDataField26":"","merchantDefinedDataField27":"","merchantDefinedDataField28":"","merchantDefinedDataField29":"","merchantDefinedDataField30":"","merchantDefinedDataField31":"","merchantDefinedDataField32":"","merchantDefinedDataField33":"","merchantDefinedDataField34":"","merchantDefinedDataField35":"","merchantDefinedDataField36":"","merchantDefinedDataField37":"","merchantDefinedDataField38":"","merchantDefinedDataField39":"","merchantDefinedDataField40":"","merchantDefinedDataField41":"","merchantDefinedDataField42":"","merchantDefinedDataField43":"","merchantDefinedDataField44":"","merchantDefinedDataField45":"","merchantDefinedDataField46":"","merchantDefinedDataField47":"","merchantDefinedDataField48":"","merchantDefinedDataField49":"","merchantDefinedDataField50":"","merchantDefinedDataField51":"","merchantDefinedDataField52":"","merchantDefinedDataField53":"","merchantDefinedDataField54":"","merchantDefinedDataField55":"","merchantDefinedDataField56":"","merchantDefinedDataField57":"","merchantDefinedDataField58":"","merchantDefinedDataField59":"","merchantDefinedDataField60":"","merchantDefinedDataField61":"","merchantDefinedDataField62":"","merchantDefinedDataField63":"","merchantDefinedDataField64":"","merchantDefinedDataField65":"","merchantDefinedDataField66":"","merchantDefinedDataField67":"","merchantDefinedDataField68":"","merchantDefinedDataField69":"","merchantDefinedDataField70":"","merchantDefinedDataField71":"","merchantDefinedDataField72":"","merchantDefinedDataField73":"","merchantDefinedDataField74":"","merchantDefinedDataField75":"","merchantDefinedDataField76":"","merchantDefinedDataField77":"","merchantDefinedDataField78":"","merchantDefinedDataField79":"","merchantDefinedDataField80":"","merchantDefinedDataField81":"","merchantDefinedDataField82":"","merchantDefinedDataField83":"","merchantDefinedDataField84":"","merchantDefinedDataField85":"","merchantDefinedDataField86":"","merchantDefinedDataField87":"","merchantDefinedDataField88":"","merchantDefinedDataField89":"","merchantDefinedDataField90":"","merchantDefinedDataField91":"","merchantDefinedDataField92":"","merchantDefinedDataField93":"","merchantDefinedDataField94":"","merchantDefinedDataField95":"","merchantDefinedDataField96":"","merchantDefinedDataField97":"","merchantDefinedDataField98":"","merchantDefinedDataField99":"","merchantDefinedDataField100":""}

Ej con formato JSON:

{
"merchantId": "**********",
"name": "**********",
"password": "********",
"mode": "AUT",
"controlNumber": "PBA22042709695XXFA36H01",
"terminalId": "12348",
"amount": "1.00",
"merchantName": "Pruebas Banorte",
"merchantCity": "Monterrey",
"lang": "ES",
"customerRef1": "22042709695XXFA36H01",
"customerRef2": "",
"customerRef3": "",
"customerRef4": "",
"customerRef5": "",
"emailCdmx": "correo@correo.group",
"phoneCdmx": "5566778899",
"conceptoCdmx": "|Concepto:|Impuestos sobre Tenencia de Veh?culos y derechos|Referencia:|P31BFM|Ejercicio Fiscal:|2021",
"billToFirstName": "Nombre Cliente",
"billToLastName": "Apellidos Cliente",
"billToStreet": "Eugenia",
"billToStreetNumber": "197",
"billToStreetNumbe2": "",
"billToStreet2Col": "Narvarte",
"billToStreet2Del": "Benito Juarez",
"billToCity": "CDMX",
"billToState": "DF",
"billToCountry": "MX",
"billToPhoneNumber": "5566778899",
"billToPostalCode": "03999",
"billToEmail": "review@banorte.com",
"billToCustomerId": "",
"billToCustomerPassword": "",
"billToDateOfBirth": "",
"billToHostname": "LSTK",
"billToHttpBrowserEmail": "",
"billToIpAddress": "127.0.0.1",
"comments": "Compra a MSI",
"shipToFirstName": "",
"shipToLastName": "",
"shipToStreetNumber": "",
"shipToStreetNumber2": "",
"shipToStreet2Col": "",
"shipToStreet2Del": "",
"shipToCity": "",
"shipToState": "",
"shipToCountry": "",
"shipToPostalCode": "",
"shipToPhoneNumber": "",
"merchantDefinedDataField3": "",

"merchantDefinedDataField4": "",
"merchantDefinedDataField5": "",
"merchantDefinedDataField8": "",
"merchantDefinedDataField6": "",
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Resultado:

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Armado de cadena final

En base al objetivo, se tiene que armar una cadena final con las dos subcadenas.

Subcadena1:::Subcadena2

Tomando los resultados de los ejemplos la cadena final quedaría:

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gVn94zyqbgqkHUAzsnTjaccIYNQ/whtyOOQnisqh/ZIEnTe3nETh3TmOf8q8PL/eLaFoRWochXrcnorHiWi4Bdrme
E5hqW3LFUnZNRbGQ+8OaU10peZYtBjNFPB5sw3CyW2j2QEo9kL4hFsK1VsHXQcdtAIXji3ZmiTtbWvEuHY+eI
JoHctcGIo4WhAfTv0EmNpMyEqn7CAfviUixJYrStQck+bWVToEyydSdYeLLHcjKn3FmX52WJuRWwrQIO/+4IIvIf
Ht/aC4Wy0giq+kDS4g9N8N2lhRTk2LbAC/xQWGmo6PN+d547bICVL2Tm7OMCC9Z0b96Q7rVFAQBgT1sku9br
ig/7GyQIqKXiPRIrD6fK1VbZ7DRkiqung3q013EJrEVqKS6gmOW5bO57LOguHv4r677Jg+Jt9aYLdg7t8kqADxf4G
LkoBvHLXVOVfFaavwOW67fTups9H55LA6TkEYS1BgJqDlfrDmMKViyzJYOWcg/1PfpsgsdcSvlW9/EKZqzwkcV
Uz9TBVrFa/uyukLog3730ifuohW9DRXQozfQSJHVaQf6n2QNTpBB8QIrRLSKSV88rN6caVBeTURzPKU3j1IdZV
evEO22+FQz7013cHOynO6qXxpuhaOP9eGoAe/3DGRAny6tXqfiyy7MB1IWCNLa4LUL6bW13nIxvpYSrvf36C4d
k2aWKaSubLUyqXzlVXCtsQyRlPrZiGMIfIAKM3WAWkzINpYxIDuG/076GNRMgzOCwp4MPXIVz4oa+Sk4o9Lie
OAhVTTeS0qfNowvyiOdxHhtJb2aBeuq/TS10TuYIIH3Dmd+/8FKyC2NvPXA6Uxk/unzKECZfJnURx62/66lI4ZCO
1I+azATD8vwgNH7fk5D2rjSFTzNuYoWrjJ599k6i59aE0kGW6gyfV2bAAU08F6My7iWOoX6zDdjNNudf4IPFljGY
bmwOXpzzx9ArzBasVy3BjUXisaDdA6tj3sz9meLCR7jqlRqj9Oqssc91vXOM9IktOc6/4qZmiKTZxcnKDwSEeGN
W4oG7k9WwLNYZ9ekUN68aGOsoKKzkcI8t3KrYReoskvt1GpZHI1qz1/8prFUX2Yyex0rs69IO4+GfYe6aNLdskU
WuHNOOvpNpk6+ha+vqfIXzNqPrFWUlluDL1cNFe8+7zzqErbsm7o/VoYnisIIHIwOcfHmvHqOjvW6MU6In4tBLgv
wPs8NrsZeCn96rAuwrdSw4M0s0ohmNB/UdXJwmymP8XNXB0XzOLSFgltJebMIPU9vjozip65tHJv4ajn/hbsga9
HN4ENmFhYZEA4/AUTKXCrNloy1pptrR2jFYh+s/PHGDAoMhUuHDnBOp+uznw4dj3y8ya3i+OrPhhPYc2ZZKOs
8gWdlOR8xKWgxMTLPHIEdA+ev31YESzKx81yXt4TnN5xHOKlazIVpG0LXA1Wn3DCEAPou5kDnGJEGRe9Mq
GotvRBBI0djEAzdhhYH54k1p9ojmP7xbyGxYJBUKk8VAXdZ/vHZx1bWTjlaN3RQYxuAsTz1mUu5Fb54WppLRN
SBS15WKT8gWKZ6zFWFb04JgRtjEo1q5GRideObSHn+HBcRtLtGYJ4lUTICVtsiOjl/OHyy+0T32y9AZTkUVQIC
Olbu48Vy1LmWWjv1MRWK0aeKb7FSjPKltX/JQHWenIZsO5Dy3QD8wOcYrbWMHH0crC589QSdqW6wAB5Bfh
D9cMhqGAGL2xi9vqdVSetjvHiGcSF0l+d+78yx9IR4i18saXrMbIgq58+kp8ww+OecTRj1l08HTAF4tFhUlkoCzZyX/
8hqdYr7DGzaoDhGftQ3Nyg4aK8pyt3YDD2JNoL/lcjQAusuhhnYM7Ooi7rxshU4fg1w31INKNODUmQmjh2+C7Bg
LIUyzrl9EFQTAvBrKRNgAJPUuCqkya0JZMiEWP0flVqYpviT5jHuNEhrvDc2D0vP4VMCFvLI5VWjSS3gUyxR8Hi
Gl77VKPfi+IK8DRVid5TA+WFUazEwgaKqBdwKHXhVdNncsrfgsxSVc4tkZOoB577VAuex2X+7yibfjeyGjMNobx3
bB5Phh9y/lf621q9sgmHMv2xp+q8kvVJ5MnLPcObJpBYHHDnzcLI0uK0Tf5V+GUSg/wEla+0FoSkF1OWe91BG
OGWcLZwY5Il9RSoUg965ZImVc

4.- Descifrado de respuesta de VCE

Para descifrar la información que está encriptada de manera simétrica, será necesario contar con los valores que se generaron de manera inicial en el cifrado AES (**viHex::saltHex::passprhase**), estos valores se sugieren que se queden a resguardo por parte del comercio, dado que son la llave que utilizará para descifrar la información que la VCE retorne:

Código:

Se instancia la clase para decodificar:

```
CifradoAES helper = CifradoAES.getInstance();
```

```
public static com.cifrado.utils.CifradoAES getInstance() {
  if (instance == null) {
    instance = new com.cifrado.utils.CifradoAES(128, 1000);
  }
  return instance;
}

private CifradoAES(int keySize, int iterationCount) {
  this.keySize = keySize;
  this.iterationCount = iterationCount;
  try {
    this.cipher = Cipher.getInstance("AES/CTR/NoPadding");
  }
  catch (NoSuchAlgorithmException|javax.crypto.NoSuchPaddingException e) {
    throw fail(e);
  }
}
```

Se invoca el método para descifrar:

```
String descifradoAES = helper.decrypt(request.getSalt(), request.getVi(), request.getPassphrase(), request.getCypherData());
```

```
public String decrypt(String salt, String iv, String passphrase, String ciphertext) {
  try {
    SecretKey key = generateKey(salt, passphrase);
    byte[] decrypted = doFinal(2, key, iv, base64(ciphertext));
    return new String(decrypted, StandardCharsets.UTF_8);
  } catch (Exception e) {
    return null;
  }
}

private byte[] doFinal(int encryptMode, SecretKey key, String iv, byte[] bytes) {
  try {
    this.cipher.init(encryptMode, key, new IvParameterSpec(hex(iv)));
    return this.cipher.doFinal(bytes);
  }
  catch (InvalidKeyException|java.security.InvalidAlgorithmParameterException|javax.crypto.IllegalBlockSizeException|javax.crypto.BadPaddingException e) {
    return null;
  }
}

private SecretKey generateKey(String salt, String passphrase) {
  try {
    SecretKeyFactory factory = SecretKeyFactory.getInstance("PBKDF2WithHmacSHA1");
    KeySpec spec = new PBEKeySpec(passphrase.toCharArray(), hex(salt), this.iterationCount, this.keySize);
    SecretKey key = new SecretKeySpec(factory.generateSecret(spec).getEncoded(), "AES");

    return key;
  }
  catch (NoSuchAlgorithmException|java.security.spec.InvalidKeySpecException e) {
    return null;
  }
}
```

```
public static byte[] hex(String str) {
  try {
    return Hex.decodeHex(str.toCharArray());
  }
  catch (DecoderException e) {
    throw new IllegalStateException(e);
  }
}
```

salt = Elemento **saltHex** generado al momento de realizar el cifrado simétrico en AES.

Ej

3076356C346F7366387675337A316171

iv = Elemento **ivHex** generado al momento de realizar el cifrado simétrico en AES.

Ej

7434316130317A753068723272746569

phassphrase = Elemento **passPhrase** generado al momento de realizar el cifrado simétrico en AES.

Ej.

g-T=Kq81GSgxRhc%

chypertext = Valor del elemento **data** del JSON de respuesta de VCE.

Ej.

{
"**data**":
"XxsiFPMZjPSOjJyanlLHARoa/+lIdegR3lUczOcD4qGa4a4InjrkeL+QY6PpGpJwDoimBOgbB/0E03YQBxNKMIR
X8fZvKxszHcsZ16IgyB/gZHhv+/sguHFMYC0Ccu5hHQejZzX0/5OmXeGkMm6XhlxLKlOhbzcNCaEOQru23snBrIf
voJN8lWC+q4FlQKIGS3TiFLGQGSBahB3ePnysGAiGD4eWrvtRmUM8NdiWJ6AEWKYFtnhAshTc3NprUIjA7IGr
qrHaOtQUw/Zrz7s5ng/Fjj9CI9bE+UyPpjGzPwGtRkkg4G1jGZWPuyJw7yH5tb/q7MboZX8zb5nbVuikDyDhQw==",
"status3D": "200",
"eci": "05",
"decision": "REVIEW",
"reasonCode": 480,
"bnteCode": "00",
"bnteText": "Transaccion Exitosa",
"id": 2,
"message": "operationSuccessfully",
"numeroControl": "PBA22042709695XXFA36H01"
}

Resultado:

{"idAfiliacion":7000002,"referencia":248233587990,"numeroControl":"PBA22042709695XXFA36H01","fechaReqCte":"2022-04-28T13:41:18.558","fechaRspCte":"2022-04-28T13:41:18.684","resultadoPayw":"A","codigoAut":"214414","texto":"Aprobada por Payworks (modo prueba)"}

Formateado como JSON:

{
"idAfiliacion": 7000002,
"referencia": 248233587990,
"numeroControl": "PBA22042709695XXFA36H01",
"fechaReqCte": "2022-04-28T13:41:18.558",
"fechaRspCte": "2022-04-28T13:41:18.684",
"resultadoPayw": "A",
"codigoAut": "214414",
"texto": "Aprobada por Payworks (modo prueba)"
}